🧾 Inventory Management App (Frontend Only)

🎯 Objective

Build a simple web app where users can:

- Add products
- View inventory
- Update quantities
- Delete items
- Persist data (using browser storage)

🧠 What Each Language Does

🟢 HTML (Structure)

Defines **what exists** on the page:

- Forms (input fields, buttons)
- Tables (to display inventory)
- Layout containers

👉 Think of HTML as the **skeleton**

🔵 CSS (Styling)

Controls **how everything looks**:

- Colors, spacing, fonts
- Layout (grid, flexbox)
- Responsive design

👉 Think of CSS as the **appearance**

🟡 JavaScript (Logic)

Handles **behavior and functionality**:

- Adding/removing items
- Updating inventory
- Saving data (localStorage)
- Dynamic updates (no page reload)

👉 Think of JS as the **brain**

🏗️ Project Structure

Create 3 files:

inventory-app/
├── index.html
├── style.css
└── script.js

🧱 Step 1: HTML (Structure)

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Inventory Manager</title>
 <link rel="stylesheet" href="style.css">
</head>
<body>

 <div class="container">
   <h1>Inventory Management</h1>

   <!-- Form -->
   <form id="inventory-form">
     <input type="text" id="name" placeholder="Product Name" required>
     <input type="number" id="quantity" placeholder="Quantity"
required>
     <input type="number" id="price" placeholder="Price" required>
     <button type="submit">Add Item</button>
   </form>
```

```
  <!-- Inventory Table -->
  <table>
    <thead>
      <tr>
        <th>Name</th>
        <th>Quantity</th>
        <th>Price</th>
        <th>Actions</th>
      </tr>
    </thead>
    <tbody id="inventory-list"></tbody>
  </table>
 </div>

 <script src="script.js"></script>
</body>
</html>
```

🎨 Step 2: CSS (Professional UI)

```
body {
 font-family: Arial, sans-serif;
 background: #f4f6f9;
 margin: 0;
}

.container {
 width: 80%;
 margin: 40px auto;
 background: white;
 padding: 20px;
 border-radius: 10px;
}

h1 {
 text-align: center;
}

form {
 display: flex;
```

```
 gap: 10px;
 margin-bottom: 20px;
}

input, button {
 padding: 10px;
 font-size: 16px;
}

button {
 cursor: pointer;
}

table {
 width: 100%;
 border-collapse: collapse;
}

th, td {
 padding: 12px;
 border-bottom: 1px solid #ddd;
 text-align: center;
}

button.delete {
 background: red;
 color: white;
 border: none;
}

button.edit {
 background: orange;
 color: white;
 border: none;
}
```

⚙️ Step 3: JavaScript (Core Logic)

```
const form = document.getElementById("inventory-form");
const list = document.getElementById("inventory-list");
```

```
let inventory = JSON.parse(localStorage.getItem("inventory")) || [];

// Render items
function renderInventory() {
 list.innerHTML = "";

 inventory.forEach((item, index) => {
   const row = document.createElement("tr");

   row.innerHTML = `
     <td>${item.name}</td>
     <td>${item.quantity}</td>
     <td>${item.price}</td>
     <td>
       <button class="edit" onclick="editItem(${index})">Edit</button>
       <button class="delete"
onclick="deleteItem(${index})">Delete</button>
     </td>
   `;

   list.appendChild(row);
 });

 localStorage.setItem("inventory", JSON.stringify(inventory));
}

// Add item
form.addEventListener("submit", function(e) {
 e.preventDefault();

 const name = document.getElementById("name").value;
 const quantity = document.getElementById("quantity").value;
 const price = document.getElementById("price").value;

 inventory.push({ name, quantity, price });

 form.reset();
 renderInventory();
});

// Delete item
function deleteItem(index) {
 inventory.splice(index, 1);
```

```
 renderInventory();
}

// Edit item
function editItem(index) {
 const item = inventory[index];

 document.getElementById("name").value = item.name;
 document.getElementById("quantity").value = item.quantity;
 document.getElementById("price").value = item.price;

 inventory.splice(index, 1);
 renderInventory();
}

// Initial render
renderInventory();
```

🚀 Key Features Implemented

✅ Add products
✅ Edit products
✅ Delete products
✅ Persistent storage (localStorage)
✅ Dynamic UI updates